HERIOT-WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

DEPARTMENT OF ACTUARIAL MATHEMATICS AND STATISTICS

F72ZA1
FINANCIAL MATHEMATICS 1

4 December 2006

Time allowed 2 hours

Attempt all **7** questions

Total Marks 60

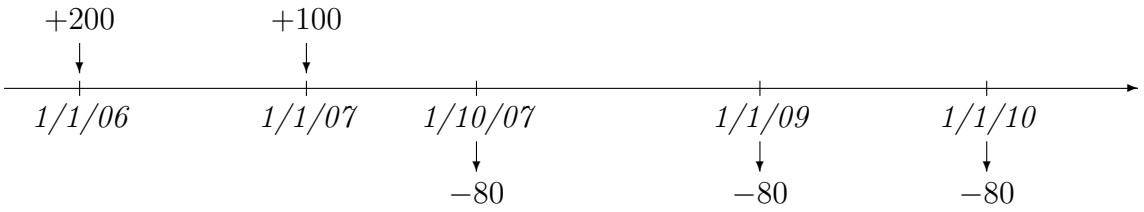
Only approved calculators may be used.

1. Fill in the missing values in the following table:

i	v	d	$i^{(4)}$	$d^{(2)}$
		0.10		

[4 marks]

2. The effective rate of interest will be 4% per half-year during 2005 and 2006, 2.5% per quarter-year during 2007, and 1% per month thereafter. A series of cashflows is given as follows:



(a) What is the present value on 1 January 2005 of these cashflows?

(b) What is the value on 1 July 2008 of these cashflows?

[7 marks]

3. (a) Show from first principles that:

$$\ddot{s}^{i}_{\overline{n}|} = \frac{(1+i)^n - 1}{d}.$$

[3 marks]

(b) Evaluate the following:

(i) $\ddot{a}^{0.07}_{\overline{10}|}$.

(ii) ${}_{10}|\ddot{a}^{0.11}_{\overline{15}|}$.

(iii) $(Ia)^{0.07}_{\overline{10}|}$.

[7 marks]

[Total 10 marks]

continued ...

4. A loan of £100,000 is to be repaid by annual repayments, made in arrear, over 20 years. The fifth and subsequent repayments are to be level, of amount £X per year. The repayments in the first, second, third and fourth years are to be £$X/5$, £$2X/5$, £$3X/5$ and £$4X/5$ respectively. The rate of interest will be 6% per year effective in the first 10 years and 7% per year effective thereafter. What is X?

[10 marks]

5. (a) You borrow £200,000 to buy a house. The loan is to be repaid over 35 years by level annual repayments, payable in arrear. The interest rate is 5% per annum effective.

   (i) What is the level annual repayment?

   (ii) What is the capital outstanding just after the 15^{th} repayment has been made?

   (iii) What is the interest content of the 16^{th} repayment?

   [5 marks]

   (b) Just after the 20^{th} repayment has been made, you ask the lender if you can repay such a proportion of the capital outstanding that the repayments over the remainder of the term will be reduced by half. The lender agrees, but stipulates that if you do this, the rate of interest for the remainder of the term will be increased by 1%, to 6% per annum effective. How much of the capital outstanding would you therefore have to repay to reduce your remaining repayments by half?

   [5 marks]

[Total 10 marks]

continued ...

6. A government issues a bond with term 25 years, to be redeemed at a premium of 2%, with a coupon of 10% per annum payable half-yearly in arrear.

   (a) Some of the bond is purchased at outset by an investor who pays income tax at 20% and capital gains tax at 30%. The investor pays a price P per £100 nominal, such that their net redemption yield is 10% per annum effective. What is P?

   [6 marks]

   (b) Immediately after payment of the 20^{th} coupon, the investor sells the bond for £90 per £100 nominal to a second investor, who pays no taxes. What redemption yield per annum does the second investor obtain?

   [6 marks]

   [Total 12 marks]

7. An investment fund has assets worth £1,000,000 on 31 December 2005. It receives an inflow of £120,000 on 1 January 2006 and pays out £250,000 on 1 January 2007. On 31 December 2007 it has assets of £900,000. Then on 1 January 2008 it pays out a further £100,000, and on 1 January 2009 it receives an inflow of £120,000. On 31 December 2009 it has assets worth £950,000.

   (a) Find the linked internal rate of return during the calendar years 2006–2009, based on the subdivision of the period into 2006–2007 and 2008–2009.

   [5 marks]

   (b) Suppose also that the fund has assets of £1,100,000 on 31 December 2006 and £850,000 on 31 December 2008. Find the time weighted rate of return during the calendar years 2006–2009.

   [2 mark]

   [Total 7 marks]

Total 60 marks